EXP NO: 19 **Experiment to understand Logistic Regression for a given data set.**

AIM:

To understand and implement Logistic Regression on a given dataset to predict categorical outcomes by establishing the relationship between input features and the target variable.

PROGRAM:

```
import numpy as np

import pandas as pd

from google.colab import files

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LogisticRegression

from sklearn.metrics import classification_report


uploaded = files.upload()

df = pd.read_csv('Social_Network_Ads.csv')


print(df.head())


features = df.iloc[:, [2, 3]].values

label = df.iloc[:, 4].values


for i in range(1, 401):

    x_train, x_test, y_train, y_test = train_test_split(

        features, label, test_size=0.2, random_state=i

    )


    model = LogisticRegression()

    model.fit(x_train, y_train)
```

```
    train_score = model.score(x_train, y_train)
    test_score = model.score(x_test, y_test)

    if test_score > train_score:
        print("Test", test_score, "Train", train_score,
              "Random State", i)

x_train, x_test, y_train, y_test = train_test_split(
    features, label, test_size=0.2, random_state=37
)

finalModel = LogisticRegression()
finalModel.fit(x_train, y_train)

print("Training Accuracy:", finalModel.score(x_train, y_train))
print("Testing Accuracy:", finalModel.score(x_test, y_test))

print(classification_report(label, finalModel.predict(features)))
```

OUTPUT:

```
    User ID  Gender  Age  EstimatedSalary  Purchased
0  15624510    Male   19            19000          0
1  15810944    Male   35            20000          0
2  15668575  Female   26            43000          0
3  15603246  Female   27            57000          0
4  15804002    Male   19            76000          0
Training Accuracy: 1.0
Testing Accuracy: 1.0
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        25
           1       1.00      1.00      1.00        18

    accuracy                           1.00        43
   macro avg       1.00      1.00      1.00        43
weighted avg       1.00      1.00      1.00        43
```

RESULT:

Thus, the Logistic Regression algorithm was successfully implemented on the given dataset to predict the target variable, and the model's performance was evaluated.